

On the Potential of Large-scale Extended Reality Interaction for Industrial Environments

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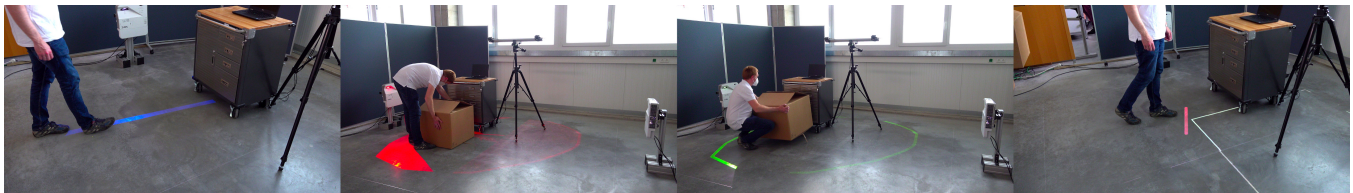


Figure 1: An overview of various use cases from left to right: (1) a guide path directs workers movements, (2) the worker is made aware of non-ergonomic behaviour and changes position accordingly (3), and a restricted area (4) warns against unintended entering.

ABSTRACT

Nowadays factories are highly flexible and adaptive industrial environments backed by digital models. To ease the interaction of human workers with these environments and represent hidden information and contexts, we propose a large-scale extended reality system for industrial settings. The huge application potential is demonstrated with various interaction methods like trigger zones, guide paths, ergonomics guidance, and restricted areas.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality.**

KEYWORDS

spatial augmented reality; pose estimation; human machine interaction

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1 INTRODUCTION

There are various ways to technically realize Extended Reality (ER). In contrast to monitors, head-mounted displays or hand-held devices, the form of Spatial Augmented Reality (SAR) utilizes digital projectors to display graphical information onto physical objects. This non-intrusive approach allows multiple persons to see the same augmentation, without the need of wearing special equipment, see Fite-Georgel [2]. Recently, interest in spatial augmentation to support workforce has increased, see Fraga-Lamas et al. [3] and Funk et al. [4]. These approaches can be considered as industrial assistive systems and are used as kind of navigation system for human-centred industrial processes. Despite their usefulness, they are limited to a workbench-sized scale due to their design.

Lately the COVID-19 pandemic has led to large-scale projection-based interaction scenarios in public places. These installations have proven their utility as distancing technology, see [6, 7]. While they are perfectly tailored to their application, they rely on expensive 3D sensing technology for providing metric interaction. Nevertheless a major impact of large-scale SAR can be expected also in other domains, e.g. industrial settings.

This work describes potential industrial use cases for a large-scale SAR projection combined with the versatile visual industrial interaction system of Heindl et al. [5], which relies on simple 2D cameras only.

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2 METHOD

Figure 2 shows an overview of the proposed system consisting of: a) an extended reality that bridges the gap between digital factory and human worker, b) pose estimation that enables the system to separate human workers from background and other objects, c) scene analysis that puts human pose and augmented reality into context and c) a freely configurable event processing component that paves the way for a multitude of applications by linking events triggered by the user with actions of the factory or feedback from the ER.

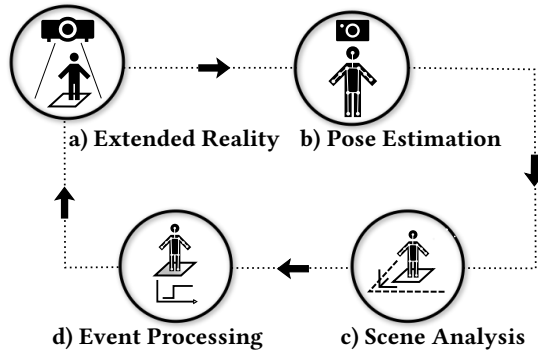


Figure 2: System Overview: a) projections on the floor for extended reality, b) pose estimation allows users to interact through their positions, c) scene analysis determines interaction of users and virtual areas, and e) event processing triggers specific events in a fictitious industrial environment, resulting in feedback to the users or other actions.

The developed SAR system consists of one camera and two ultra-short-throw projectors, see Figure 3. A checkerboard pattern, onto which gray-code patterns are projected and decoded by the camera, is used to treat the projectors as inverse cameras and calibrate the system devices' intrinsics and extrinsics. To increase robustness of the calibration process, a RANSAC scheme is applied that discards ill-fitting calibration acquisitions. A homography to the floor for each projector is used to render the projections.

Human poses are estimated with Cao et al. [1] and from each pose-skeleton the center of gravity is calculated, resulting in a floor position per user. This floor position is used for interaction with virtual zones or boundaries. The system can easily be extended with additional projectors and cameras, to enlarge the extended reality space as needed.

3 DEMONSTRATION

To demonstrate the versatility of the proposed system we consider four use cases (UCs):

- UC1** shows dynamic trigger zones projected onto the floor, see Figure 3, which are only activated by human presence and not by objects. Actuation of events can be configured freely in downstream systems, e.g. start/stop of machines.
- UC2** demonstrates the guidance of users along a guide path, that is generated on demand in response to a user query, see Figure 1. The path is projected, starting at the user's

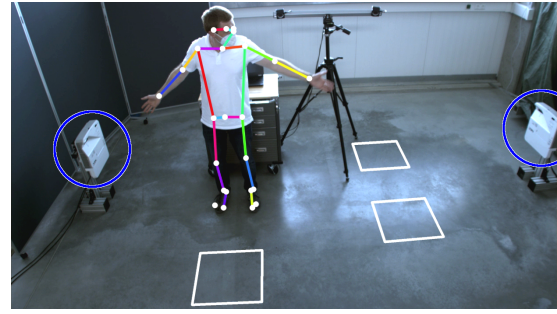


Figure 3: System setup. The system consists of one camera, whose field of view is shown in this figure, and two ultra-short-throw projectors (blue circles). The camera is used to estimate human poses. Cameras and projectors are calibrated to the floor to form a world coordinate system. This enables SAR projections that users can interact with.

feet and ending at the desired destination. Its color changes dependent on the remaining distance to the target.

UC3 illustrates ergonomics guidance, to help users avoiding injury and unsafe situations in predefined areas, see Figure 1. The border of the area changes its color to green, as soon as presence is detected. If the system is aware of non-ergonomic or dangerous postures inside the area, the color of the whole area changes to bright red and flickers (actual detection is simulated by manually triggering the system).

UC4 showcases restricted areas, see Figure 1. The user is warned of restricted areas before intruding them, by a stopline projected onto the shortest path of the user to the area. The line changes color depending on distance to the area.

4 CONCLUSION

In this work a robust and versatile spatial augmented reality system consisting of one camera and two ultra-short-throw projectors is presented. The system can be extended freely with additional devices and is applicable for a multitude of interaction use cases. The demonstrated use cases range from interactive trigger zones to guide paths that provide position aware way finding support. This demonstration should lead to fruitful discussions about further improvements and interaction possibilities for industrial settings.

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REFERENCES

- [1] Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh. 2019. OpenPose: realtime multi-person 2D pose estimation using Part Affinity Fields. *IEEE transactions on pattern analysis and machine intelligence* 43, 1 (2019), 172–186.
- [2] Pierre Fite-Georgel. 2011. Is there a reality in Industrial Augmented Reality?. In *2011 10th IEEE International Symposium on Mixed and Augmented Reality*. 201–210. <https://doi.org/10.1109/ISMAR.2011.6092387>
- [3] Paula Fraga-Lamas, Tiago M. Fernández-Caramés, Óscar Blanco-Novoa, and Miguel Vilar-Montesinos. 2018. A Review on Industrial Augmented Reality Systems for the Industry 4.0 Shipyard. *IEEE Access* 6 (2018), 13358–13375. <https://doi.org/10.1109/ACCESS.2018.2808326> Conference Name: IEEE Access.

- [4] Markus Funk, Thomas Kosch, Romina Kettner, Oliver Korn, and Albrecht Schmidt. 2016. motionEAP: An Overview of 4 Years of Combining Industrial Assembly with Augmented Reality for Industry 4.0. (2016), 4.
- [5] Christoph Heindl, Gernot Stübl, Thomas Pönitz, Andreas Pichler, and Josef Scharinger. 2019. Visual large-scale industrial interaction processing. In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers (UbiComp/ISWC '19 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 280–283. <https://doi.org/10.1145/3341162.3343769>
- [6] Hitachi Ltd. 2020. Putting fun into maintaining physical distance by system to link walking people and spatial distance : Research & Development : Hitachi. <https://www.hitachi.com/rd/news/topics/2020/1130.html>
- [7] QueueSight. 2020. Social Distancing Tool. <https://www.queuesight.com>